



CSE370: Database Management System Assignment on Functional Requirements

Proposed Project Title: Project Title

Group No: 08, CSE370 Lab Section: 11, Summer 2026		
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Submission Date:27/06/2026

Project Overview

The **Disaster Relief Management System** is a platform designed for efficient management of relief resources, volunteers, and disaster zones. This system enables administrators to track warehouse inventories, relief shipments(inventory items), oversee the areas which are affected by disasters and supervise volunteers. It keeps a record of relevant data for each disaster zone, which includes location, severity and current state, so that admins can allocate volunteers and resources more efficiently. Moreover the platform provides secure authentication for all users, who are either admins or customers, ensuring that all the users are recognized by the platform

and that they can access all the relevant features. The system also enables automated inventory monitoring along with low-stock alerts and expiry date review that help maintain the availability and quality of the relief supplies. Additionally, volunteers are tracked to be matched and deployed, depending on their availability, to improve the efficiency of emergency response operations. Overall, the system aims to improve coordination and timely delivery of aid and supplies during disasters.

Tech Stack:

- · Language: python, HTML, CSS, SQL, Python
- · Frontend Framework: HTML with Jinja2 template
- · Backend Framework: Flask
- · Styling: CSS
- · Database: MySQL
- · ODM: N/A
- · Deployment: LocalHost/XAMPP
- · External APIs:N/A

Functional Requirements

Module 1	
Shadeed	Feature Description
1	Login: Customers can log in using proper credentials or admin can login based on the specific credentials mentioned in the admin table in the database for the system.
2	Volunteer Application and Viewing Active Disaster Zones: Customers can view current active disaster zones and also choose to become a volunteer by submitting the form with providing proper information required.

3	Approve or Delete Volunteers: Admin can approve volunteers after their form has been successfully submitted. They can also choose to delete volunteers.
4	Deploy/Release Volunteers: Admin can deploy volunteers by seeing their specialty and assign them roles for a certain disaster zone. Their time of dispatch is stored. They can be released from deployment, upon which their total hours deployed is stored.
5	Volunteer Availability Toggle: Volunteers can toggle between “Available” and “Unavailable”. If they are unavailable, they cannot be deployed.

Module 2	
Member	Feature Description
Aryan	Disaster Zone & Shipment Tracking: allows administrators to declare, monitor, and update active disaster zones. The system records crucial information such as the crisis name, location, severity level, and the assigned source warehouse. Administrators can also update the operational status of these zones to track the progress of relief shipments from pending to resolved, ensuring high-severity zones are prioritized. Main functions: ● Declare and register new disaster zones ● View all active disaster zones ordered by severity ● Assign a specific source warehouse to a disaster zone ● Update the shipment and operational status of disaster zones

Aryan	Relief Shipment Dispatcher: connects inventory warehouses to active disaster zones by allowing administrators to dispatch relief supplies. The system checks warehouse availability, dynamically deducts dispatched quantities from inventory, records a permanent itemized shipment log, and automatically updates the disaster zone's status to 'Dispatched'. Main functions: • View available items in the assigned warehouse • Validate stock quantities before dispatch • Deduct inventory stock dynamically • Record each dispatch event (item, zone, quantity, admin, timestamp) in a persistent shipment log • Auto-update disaster zone status
Aryan	Live Disaster Zone Analytics & Resource Report: provides administrators with a real-time analytical overview of all crisis zones: active volunteer headcount, live accumulated hours, and total relief supplies dispatched, all prioritized by severity. Main functions: • View centralized analytics for all disaster zones • Count active personnel deployed per zone • Calculate live running totals of volunteer hours • Summarize total supplies dispatched per zone • Order and prioritize zones by severity level

Module 3	
Member	Feature Description

Oishee	Warehouse Management & Inventory Assignment: allows administrators to register and manage disaster relief warehouses. Administrators can view warehouse information and assign inventory items to specific warehouses, including their shelf locations. This helps track which items are stored in each warehouse and where they are located, making inventory organization and relief distribution more efficient. Main functions: ● Register and manage warehouses ● View warehouse information ● Assign inventory items to warehouses ● Record shelf locations ● View items stored in each warehouse
Oishee	Inventory Stock Monitoring & Low-Stock Management: provides administrators with a centralized view of inventory items and their current stock levels. The system displays information such as item names, categories, and quantities, allowing administrators to monitor available relief supplies. Items with low quantities can be identified for restocking, helping ensure that essential supplies remain available during disaster response. Main functions: ● View all inventory items ● Monitor current quantities and categories ● Identify items requiring restocking ● Update inventory stock levels ● Add or remove stock when necessary
Oishee	Expiry Date Auditor: monitors the expiration dates of relief supplies and calculates the number of days remaining before each item expires. Items that are close to expiry or already expired are clearly flagged, allowing administrators to prioritize their distribution or remove and replace unusable

	supplies.
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Main functions:

- Monitor inventory expiration dates
- Calculate days remaining until expiry
- Flag urgent and expired items
- Help prioritize items for distribution
- Support timely replacement of expired supplies